

Problem 3

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Problem 1 Find the x -coordinates of global extrema and global extreme values of f on the given interval.

(a) $f(x) = x\sqrt{2-x^2}$ on $[-\sqrt{2}, \sqrt{2}]$.

Explanation. First note that f is continuous on this closed interval so, by the Extreme Value Theorem, it must attain a maximum and minimum value on the interval $[-\sqrt{2}, \sqrt{2}]$. Those global extrema must occur either at critical points or at the boundary points.

$$\begin{aligned} f'(x) &= \sqrt{2-x^2} + x \left(\frac{1}{2\sqrt{2-x^2}}(-2x) \right) \\ &= \sqrt{2-x^2} - \frac{x^2}{\sqrt{2-x^2}} \\ &= \frac{2-x^2-x^2}{\sqrt{2-x^2}} \\ &= \frac{2(1-x^2)}{\sqrt{2-x^2}} \end{aligned}$$

Critical points of f occur where $f'(x) = 0$ or where $f'(x)$ does not exist. Solving $f'(x) = 0$ yields that $2(1-x^2) = 0$, or $x = \pm 1$. $f'(x)$ does not exist when $2-x^2 \leq 0$. But since we are restricting to the interval $[-\sqrt{2}, \sqrt{2}]$, the only points where $f'(x)$ does not exist are the endpoints $\pm\sqrt{2}$.

The critical points of f in the interval $[-\sqrt{2}, \sqrt{2}]$ are $x = \pm 1$. (The points $x = \pm\sqrt{2}$ technically are not critical points since they are the endpoints of the domain of f .) The function f must attain its maximum and minimum values in this set $x = \{\pm 1, \pm\sqrt{2}\}$. We compute

$$\begin{aligned} f(-\sqrt{2}) &= 0 \\ f(\sqrt{2}) &= 0 \\ f(1) &= 1\sqrt{2-1} = 1 \\ f(-1) &= -1\sqrt{2-(-1)^2} = -1 \end{aligned}$$

That means f has a maximum value of 1 (at $x = 1$) and a minimum value of -1 (at $x = -1$) over the interval $[-\sqrt{2}, \sqrt{2}]$.

(b) $f(x) = x^3e^{-x}$ on $[-1, 5]$.

Explanation. First note that f is continuous on this closed interval so, by the Extreme Value Theorem, it must attain a maximum and minimum value on the interval $[-1, 5]$. Those global extrema must occur either at critical points or at the boundary points.

$$\begin{aligned} f'(x) &= 3x^2e^{-x} + x^3(-e^{-x}) \\ &= x^2e^{-x}(3 - x) \end{aligned}$$

Notice that $f'(x)$ always exists so any critical point of must f occur when $f'(x) = 0$. Solving this equation:

$$\begin{aligned} x^2e^{-x}(3 - x) &= 0 \\ x^2(3 - x) &= 0 \\ x = 0 \quad \text{or} \quad x &= 3 \end{aligned}$$

Since both critical points are in the given interval, the global extrema must occur at the points $x = -1, 0, 3, 5$.

$$\begin{aligned} f(-1) &= -e \\ f(0) &= 0 \\ f(3) &= 27e^{-3} \\ f(5) &= 125e^{-5} \end{aligned}$$

Since $-e$ is the only negative value, the minimum value of f over the interval is $-e$. Since $e^3 < 27$, $27e^{-3} > 1$. But $e^5 > 125$, and so $125e^{-5} < 1$. Thus the maximum value of f over the interval is $27e^{-3}$.

(c) $f(x) = x \ln\left(\frac{x}{5}\right)$ on $[0.1, 5]$.

Explanation. First note that f is continuous on this closed interval so, by the Extreme Value Theorem, it must attain a maximum and minimum value on the interval $[0.1, 5]$.

$$\begin{aligned} f'(x) &= \ln\left(\frac{x}{5}\right) + x \cdot \frac{5}{x} \cdot \frac{1}{5} \\ &= \ln\left(\frac{x}{5}\right) + 1 \end{aligned}$$

Notice that $f'(x)$ exists for all values in $[0.1, 5]$, and so all of the critical points of f occur when $f'(x) = 0$. Solving this equation:

$$\begin{aligned} \ln\left(\frac{x}{5}\right) + 1 &= 0 \\ \ln\left(\frac{x}{5}\right) &= -1 \end{aligned}$$

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$$\begin{aligned}\frac{x}{5} &= e^{-1} \\ x &= 5e^{-1} = \frac{5}{e}\end{aligned}$$

Since $1 < \frac{5}{e} < 2$, $\frac{5}{e}$ is in the given interval. So we need to consider the points $x = \frac{1}{10}, \frac{5}{e}, 5$.

$$f\left(\frac{1}{10}\right) = \frac{1}{10} \ln\left(\frac{1}{50}\right) = \frac{1}{10} (\ln 1 - \ln 50) = -\frac{1}{10} \ln 50$$

$$f\left(\frac{5}{e}\right) = \frac{5}{e} \ln\left(\frac{1}{e}\right) = -\frac{5}{e}$$

$$f(5) = 5 \ln 1 = 0$$

Since the first two values are negative, 0 is the maximum value of f on $[0.1, 5]$. Also, since $e^{10} > 50$, $\ln 50 < 10$ and therefore $-1 < -\frac{1}{10} \ln 50$. But clearly $-\frac{5}{e} < -1$, and so $-\frac{5}{e}$ is the minimum value of f on $[0.1, 5]$.